

UNIT 3

Concept Learning

Study Guide

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1. Introduction to Concept Learning

Concept Learning:

Concept Learning is a supervised learning task in which a machine learns a concept from labeled training examples.

The objective is to:

- Identify patterns from data
- Predict unseen examples
- Build hypotheses representing the target concept

Example

A system learns whether a day is suitable for playing tennis based on:

- Weather
- Temperature
- Humidity
- Wind

2. Formulation of Hypothesis

What is a Hypothesis?

A hypothesis is a statement or rule used by a machine learning model to classify data.

A hypothesis is a function:

$$h: X \rightarrow \{0,1\}$$

where:

X = instance space

0 or 1 = class labels

Hypothesis Space (H) = The set of all possible hypotheses is called the hypothesis space.

Example

Attributes:

Sky = {Sunny, Rainy}

Wind = {Strong, Weak}

Possible hypotheses:

- Sunny AND Strong
- Rainy AND Weak
- Any combination

Types of Hypotheses

- 1) Specific Hypothesis
- 2) General Hypothesis
- 3) Simple Hypothesis
- 4) Complex Hypothesis

1. Simple Hypothesis

A hypothesis that predicts an outcome using a simple condition or a single attribute.

Example

If:

Sky = Sunny

Then:

Enjoy Sport = Yes

This uses only one condition.

2. Complex Hypothesis

A hypothesis that uses multiple conditions or combinations of attributes.

Example

If:

Sky = Sunny

Wind = Strong

Humidity = Normal

Then:

Enjoy Sport = Yes

This contains multiple conditions.

3. Specific Hypothesis

A hypothesis with very strict conditions.

It covers fewer examples.

Example

(Sunny, Warm, Normal, Strong)

Only exact matches are accepted.

4. General Hypothesis

A hypothesis with relaxed conditions using “?” symbols.

It covers many examples.

Example

(Sunny, ?, ?, Strong)

This accepts:

Any temperature

Any humidity

as long as:

Sky = Sunny

Wind = Strong

3. Version Space

Version Space is the subset of hypotheses consistent with all training examples.

Formula

$$VS_{H,D} = \{h \in H | \text{Consistent}(h,D)\}$$

Where:

H = hypothesis space

D = training dataset

Boundaries of Version Space

- 1) Specific Boundary (S): Most specific hypothesis
- 2) General Boundary (G): Most general hypothesis

4. Hypothesis Elimination

Hypothesis Elimination removes inconsistent hypotheses from the hypothesis space.

Goal

Keep only those hypotheses that correctly classify training examples.

Working Principle

Positive examples generalize S

Negative examples specialize G

5. Candidate Elimination Algorithm

Candidate Elimination Algorithm incrementally updates:

Specific boundary (S)

General boundary (G)

to identify the correct version space.

Algorithm Steps

Step 1: Initialize:

$S = \langle \phi, \phi, \phi \rangle$

$G = \langle ?, ?, ? \rangle$

Step 2:

For each positive example:

Remove inconsistent hypotheses from G

Generalize S minimally

Step 3:

For each negative example:

Remove inconsistent hypotheses from S

Specialize G minimally

Step 4:

Repeat until all examples are processed.

6. Example of Candidate Elimination

Example

Example	Sky	AirTemp	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Strong	No

Initial Boundaries

$S = \langle \phi, \phi, \phi, \phi \rangle$

$G = \langle ?, ?, ?, ? \rangle$

After Positive Examples: S becomes generalized.

After Negative Examples: G becomes specialized.

7. Probabilistic Approximately Correct (PAC) Learning

Probably Approximately Correct Learning PAC Learning explains whether a machine can learn a concept:

- Approximately correct
- With high probability
- Using finite examples

PAC Learning Parameters

- 1) Accuracy (ϵ) : Allowed error.
- 2) Confidence (δ) : Probability that the hypothesis is correct.

PAC Condition

A learner is PAC if:

$$P(\text{error}(h) \leq \epsilon) \geq 1 - \delta$$

Where:

ϵ = error tolerance

δ = confidence parameter

8. VC Dimension

Vapnik–Chervonenkis Dimension VC Dimension measures the learning capacity of a model.

The maximum number of points that can be shattered by a hypothesis class.

Shattering : A set of points is shattered if all possible classifications can be represented.

Example

A line in 2D can shatter 3 points.

Therefore:

$VC(H)=3$

Importance of VC Dimension:

- Measures model complexity
- Helps avoid overfitting
- Used in statistical learning theory
- Important for generalization analysis

9. Advantages of Concept Learning

- Improves prediction accuracy
- Learns from examples

- Useful in classification tasks
- Foundation for supervised learning

10. Limitations

- Sensitive to noisy data
- Large hypothesis space increases complexity
- Computationally expensive
- Requires labeled training data

11. Applications

- Email spam detection
- Medical diagnosis
- Recommendation systems
- Weather prediction
- Image classification

REFERENCES

1. Tom M. Mitchell Mitchell, Tom M. Machine Learning. McGraw-Hill Education, 1997.
2. Leslie Valiant Valiant, Leslie G. "A Theory of the Learnable." Communications of the ACM, vol. 27, no. 11, 1984, pp. 1134–1142.
3. Vladimir Vapnik Vapnik, Vladimir N., and Alexey Ya. Chervonenkis. "On the Uniform Convergence of Relative Frequencies of Events to Their Probabilities." Theory of Probability & Its Applications, vol. 16, no. 2, 1971, pp. 264–280.
4. GeeksforGeeks. "Candidate Elimination Algorithm in Machine Learning." Accessed 21 May 2026.
5. TutorialsPoint. "Concept Learning and Candidate Elimination Algorithm." Accessed 21 May 2026.

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